

A. 1D Eraser

8
6 3
WBWWWB 2
7 3
WWBWBWW 1
5 4
BWBWB 2
5 5
BBBBB 1
8 2
BWBWBWB 4
10 2
WBWBWBWBW 3
4 1
BBBB 4
3 2
WWW 0

Input: Such that a stripe of paper s with n cells with 2 possibilities: Black (B) or White (W).



You have the number of cells (n) and the number of consecutive operations that converts (B $\rightarrow$ W).

Output: Minimum number of operations

ALGO(s : string, n : int, k : int):

contador = 0 // Acumula los pasos para que toda la cadena sea blanca

Para $i = 0 \rightarrow n-1$:

Si $s[i]$ es Negro (B)

contador++ // Convertir la celda actual a Blanco (W)

$i = i + k - 1$ // convertir los siguientes $k - 1$, $1 + k - 1 = k$ celdas

return contador

B. Even Odds

10 3 $n = 10, k = 3$
5
7 7 $n = 7, k = 7$
6
[1 3 5 7 9] [2 4 6 8 10] $k = 3$
[1 3 5 7] [2 4 6] $k = 7$

Input: Such that two integers, last number n and the position k we are searching, also the indexes of k following $1 \leq k \leq n$.

(eg. $k = 3$ is the third element, and its index is $k-1 = 2$).

Output: The number at that position in the complete sequence of ($a_1 a_2 \dots a_{n/2} b_1 b_2 \dots b_{n/2}$)

IDEA 1: Notice that for the odd integers, there are exactly $\text{ceil } n/2$, and floor $n/2$ for even integers. Furthermore, we can "split" in 2 different subsequences ordered, so we can try parallel binary search or two pointers strategy

NO effective

IDEA 2:

if the position number $k \leq \text{count of odd numbers } (\text{ceil } n/2)$

$\Rightarrow 2*k-1 \Rightarrow \text{ALWAYS ODD}$

Otherwise (position number $k > \text{count of odd numbers } (\text{ceil } n/2)$)

$\Rightarrow 2*(k - \text{ceil}(n/2)) \Rightarrow \text{ALWAYS EVEN}$

Also we can use $(n+1)/2$

ALGO(n : int, k : int)

si $k \leq (n+1)/2$

retornar $2*k-1$

sino

retornar $2*(k - (n+1)/2)$

=====> $2*k - (n+1)$